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**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Third Semester B.Tech (CBCS) Degree Examinations**

December 2025

CV2001-1 – BUILDING MATERIALS AND CONSTRUCTION

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

1. Jumper is a tool used for  
A) Dressing of stones  
B) Preservation of stones  
C) Testing of stones  
D) Quarrying of stones
2. Which type of rock is formed from the cooling and solidification of magma  
A) Metamorphic  
B) Organic  
C) Igneous  
D) Metamorphic
3. The addition of magnesia in brick manufacturing helps in  
A) Enhancing the color of the bricks  
B) Increasing the hardness of bricks  
C) Reducing the shrinkage during drying and firing  
D) Providing resistance to cracking
4. In the brick manufacturing process, tempering refers to:  
A) Heating the bricks to a high temperature  
B) Mixing and kneading the clay to make it plastic  
C) Cooling the bricks after firing  
D) Applying a surface coating to the bricks
5. Which part of the timber cross-section is responsible for the transport of nutrients and water in a living tree  
A) Sapwood  
B) Heartwood  
C) Cambium layer  
D) Pith
6. A longitudinal crack along the growth rings in timber is known as  
A) Cup  
B) Twist  
C) Knot  
D) Shake
7. The primary advantage of plywood over solid wood is  
A) Resistance to warping and splitting  
B) Lower cost  
C) Light weight  
D) Increased hardness
8. Mild steel is primarily composed of  
A) Iron and nickel  
B) Iron and a high percentage of carbon  
C) Iron and a low percentage of carbon  
D) Iron and chromium
9. What is the primary purpose of site exploration before the construction of a foundation  
A) To assess the soil's bearing capacity and identify any potential geotechnical issues  
B) To select the building materials based on aesthetics  
C) To estimate the construction cost of the building  
D) To determine the architectural style of the building
10. The size of the plate used in a plate load test is usually  
A) Equal to the size of the actual foundation  
B) Smaller than the actual foundation  
C) Larger than the actual foundation  
D) Size does not matter in the test
11. Which of the following is classified as a shallow foundation  
A) Pile Foundation  
B) Caisson Foundation  
C) Raft Foundation  
D) Pier Foundation

12. Which of the following is NOT a shape commonly used for combined footings  
 A) Triangular B) Trapezoidal  
 C) Rectangular D) Circular
13. ....is the term for a brick laid with its shortest end exposed on the face of the wall  
 A) Soldier B) Queen closer  
 C) Stretcher D) Header
14. What is a "Queen closer" in brick masonry  
 A) A brick used to complete a course B) A brick cut diagonally to form a triangle  
 C) A brick cut in half along its length D) A full brick used at the corners
15. What is the term for the vertical support on either side of an arch  
 A) Keystone B) Abutment  
 C) Pier D) Springing line
16. The primary function of a chejja is to  
 A) Protect walls and windows from rain and sunlight B) To provide shelter or cover over an entrance or outdoor area  
 C) Offer additional floor space D) Act as a structural support for beams
17. ....is the primary purpose of a subfloor in construction  
 A) To provide an aesthetic finish B) To serve as a base for the final flooring material  
 C) To insulate the building from the ground D) To support the structural walls
18. ....is the term for the overhanging edge of a tread that extends beyond the riser  
 A) Baluster B) Newel  
 C) Nosing D) Stringer
19. Which material is most commonly used for the construction of collapsible doors  
 A) Steel B) Wood  
 C) Glass D) Plastic
20. A volatile substance added to a paint to make its application easy and smooth is known as  
 A) Base B) Additive  
 C) Solvent D) Vehicle

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I  |                                                                                                                                                                                     | Marks | BT* | CO* | PO* |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.        | a) List any Three properties of an Ideal preservative used for the preservation of stones. Explain any two methods of preserving stones.                                            | 8     | L*2 | 1   | 1   |
|           | b) Explain in detail any four defects due to Natural forces in Timber.                                                                                                              | 8     | L2  | 2   | 1   |
| 2.        | a) With the help a neat sketch explain the Burning process of bricks in Clamps.                                                                                                     | 8     | L2  | 1   | 1   |
|           | b) i) Differentiate between Ply wood and Wood board.<br>ii) Differentiate between Mild Steel and Cast Iron.                                                                         | 8     | L2  | 2   | 1   |
| 3.        | a) i) List any five characteristics of a good roof tile.<br>ii) Compare between "Class AA" and "Class A" with respect to Mangalore tiles.                                           | 8     | L1  | 1   | 1   |
|           | b) What are Bogue's compound. Explain their role in setting and hardening process of cement.                                                                                        | 8     | L2  | 2   | 1   |
| Unit – II |                                                                                                                                                                                     |       |     |     |     |
| 4.        | a) i).What is Bearing capacity of Soil? List any 4 methods adopted for increasing the bearing capacity of soil.<br>ii).List any three circumstances which requires pile foundation. | 8     | L1  | 3   | 1   |

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|                   |                                                                                                                                                                 |   |    |   |   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| b)                | Sketch the elevation of Flemish bond and mention any Five essential features of it.                                                                             | 8 | L2 | 4 | 1 |
| 5. a)             | Using Proportional sketches, Explain "Raft Foundation" and "Pile Foundation".                                                                                   | 8 | L2 | 3 | 1 |
| b)                | i) List any four rules to be adopted in order to achieve good bond in brick masonry.<br>ii) With the help a neat sketch explain Cramped Joint in Stone Masonry. | 8 | L2 | 4 | 1 |
| 6. a)             | Explain the following with the help a neat sketch<br>i) Isolated Footing ii) Strap Footing                                                                      | 8 | L2 | 3 | 1 |
| b)                | Explain with the help of a neat sketch R.C.C Lintel and Stone Lintel                                                                                            | 8 | L2 | 4 | 1 |
| <b>Unit – III</b> |                                                                                                                                                                 |   |    |   |   |
| 7. a)             | Define roof and with the help of a neat sketch explain Lean to Roof.                                                                                            | 8 | L2 | 5 | 1 |
| b)                | Explain Briefly Lath and Stucco plastering.                                                                                                                     | 8 | L2 | 5 | 1 |
| 8. a)             | Define and show (sketch) the following components of the staircase.<br>i) Nosing ii) Landing iii) Scotia iv) Tread                                              | 8 | L1 | 5 | 1 |
| b)                | Name the constituents of Distemper. Explain the step by step process of Distemping.                                                                             | 8 | L2 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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